

5.5 Equilibria (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.5 Equilibria (A Level Only)
Difficulty	Hard

Time allowed: 60

Score: /43

Percentage: /100

Question 1a

3-dodecylbenzenesulfonic acid can be prepared from benzenesulfonic acid.



Suggest the reagents and conditions and name the mechanism for this reaction.

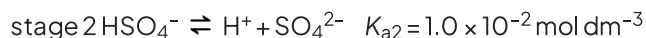
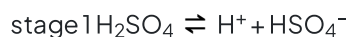
reagents and conditions

mechanism

[2 marks]

Question 1b

When concentrated sulfuric acid is added to water, dissociation takes place in two stages.



K_{a2} is the acid dissociation constant for stage 2.

i)

Write the expression for the acid dissociation constant K_{a2} .

$K_{a2} =$

[1]

ii)

H_2SO_4 is considered a strong acid whereas HSO_4^- is considered a weak acid.

Suggest how the magnitude of the acid dissociation constant for stage 1 compares to K_{a2} .

[1]

[2 marks]

Question 1c

Benzoic acid, $\text{C}_6\text{H}_5\text{CO}_2\text{H}$, is a weak acid.

The value of the acid dissociation constant, K_a , of benzoic acid is $6.3 \times 10^{-5} \text{ mol dm}^{-3}$ at 298K.

Calculate the pH of the 0.50 g dissolved in 250 cm^3 of water.

Give your answer to **two** decimal places and show your working.

pH =

[4 marks]

Question 1d

Calculate the pH of the solution formed when 0.060 g of sodium hydroxide is added to 500.00 cm^3 of $0.005 \text{ mol dm}^{-3}$ of benzoic acid. Give your answer to **two** decimal places. Show your working.

pH =

[5 marks]

Question 2a

One atom of each of the four elements H, C, N and O can bond together in different ways.

Two examples are molecules of cyanic acid, HOCN, and isocyanic acid, HNCO. The atoms are bonded in the order they are written.

i)

Draw 'dot-and-cross' diagrams of these two acids, showing outer shell electrons only.

HOCN, cyanic acid
HNCO, isocyanic acid

[3]

ii)

Suggest the values of the bond angles HNC and NCO in isocyanic acid.

HNC

NCO

[1]

iii)

Suggest which acid, cyanic or isocyanic, will have the **shorter** C–N bond length. Explain your answer.

[1]

[5 marks]

Question 2b

Isocyanic acid is a weak acid.



i)

Calculate the pH of a 0.10 mol dm^{-3} solution of isocyanic acid and show your working.

pH =
[2]

Sodium cyanate, NaNCO , is used in the production of isocyanic acid. Sodium cyanate is prepared commercially by reacting urea, $(\text{NH}_2)_2\text{CO}$, with sodium carbonate. Other products in this reaction include ammonia and steam.

ii)

Write an equation for the production of NaNCO by this method.

[1]

[3 marks]

Question 2c

Barium hydroxide, $\text{Ba}(\text{OH})_2$, is completely ionised in aqueous solutions. During the addition of 30.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ $\text{Ba}(\text{OH})_2$ to 20.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ isocyanic acid, the pH was measured.

i)

Calculate the $[\text{OH}^-]$ at the end of the addition.

$[\text{OH}^-] = \dots\dots\dots \text{mol dm}^{-3}$
[2]

ii)

Use your value in (i) to calculate $[\text{H}^+]$ and the pH of the solution at the end of the addition.

final $[\text{H}^+] = \dots\dots\dots \text{mol dm}^{-3}$
final pH = $\dots\dots\dots$
[2]

iii)

On the following axes in Fig. 2.1, sketch how the pH changes during the addition of a total of 30.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ $\text{Ba}(\text{OH})_2$ to 20.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ isocyanic acid.

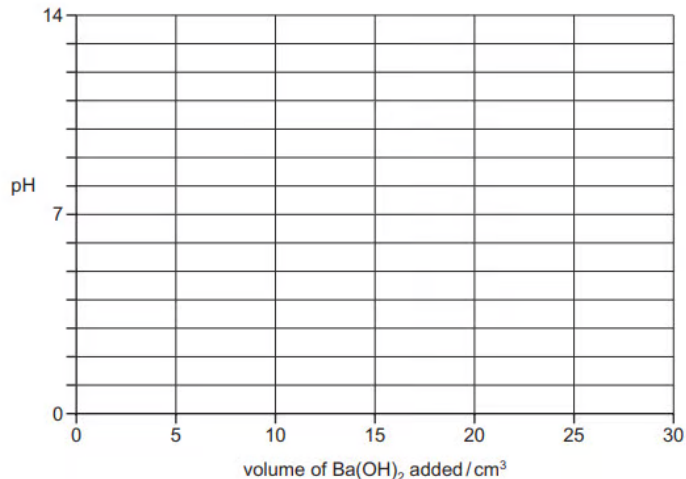


Fig. 2.1

[3]

[7 marks]

Question 3a

This question is about buffer solutions

Calculate the pH of a buffer solution made by dissolving 0.275 g of benzoic acid ($K_a = 6.3 \times 10^{-5} \text{ mol dm}^{-3}$) and 0.525 g of sodium benzoate in 250 cm^3 of water. Give your answer to 2 decimal places and show your working.

pH =

[5 marks]

Question 3b

Calculate the pH of a buffer solution made by mixing together 130 cm^3 of 0.2 mol dm^{-3} ethanoic acid ($K_a = 1.74 \times 10^{-5}\text{ mol dm}^{-3}$) and 85 cm^3 of 0.45 mol dm^{-3} sodium ethanoate. Give your answer to 2 decimal places and show your working.

pH =

[3 marks]**Question 3c**

Calculate the new pH of the buffer solution given in part **(b)** if 1.00 cm^3 of 1.00 mol dm^{-3} sodium hydroxide is added. Give your answer to two decimal places and show your working.

Comment on the pH value after the addition of sodium hydroxide.

pH =

[5 marks]

Question 3d

Write two equations for the separate addition of a small volume of base and acid to the buffer solution given in part (b).

equation 1

equation 2

[2 marks]

Model Answers: Hard

1a

a) The reagents, conditions and mechanism are:

- $\text{C}_{12}\text{H}_{25}\text{Br}$ and halogen carrier e.g. AlBr_3 (+ heat)

OR

$\text{C}_{12}\text{H}_{25}\text{Cl}$ and halogen carrier e.g. AlCl_3 (+ heat); [1 mark]

- Electrophilic substitution; [1 mark]

[Total: 2 mark]

- The change is the substitution of the $\text{C}_{12}\text{H}_{25}$ group onto the benzene ring
 - Therefore, alkylation via electrophilic substitution is required
- In order to generate the electrophile, a halogenoalkane and halogen carrier are required

1b

b)

i) The K_a expression for stage 2 is:

- $K_{a2} = \frac{[\text{H}^+][\text{SO}_4^{2-}]}{[\text{HSO}_4^-]}$; [1 mark]

ii) The difference in magnitudes are:

- K_a of H_2SO_4 is larger / greater than K_{a2} ; [1 mark]

[Total: 2 marks]

- **Remember:** To write the expression for K_a for stage 2, use $K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$
- The K_a value is the extent to which the acid dissociates
 - The stronger the acid, the higher the K_a value

1c

c) To calculate the pH of benzoic acid:

- Moles of $\text{C}_6\text{H}_5\text{COOH} = \frac{0.50}{122.0} = 0.004 \text{ mol}$

AND

Concentration of $\text{C}_6\text{H}_5\text{COOH} = 0.004 \times 4 = 0.016 \text{ mol dm}^{-3}$; [1 mark]

- $K_a = \frac{[\text{H}^+][\text{C}_6\text{H}_5\text{COO}^-]}{[\text{C}_6\text{H}_5\text{COOH}]}$

OR

$$= \frac{[\text{H}^+]^2}{[\text{C}_6\text{H}_5\text{COOH}]}$$

OR

$$= \frac{[\text{H}_3\text{O}^+][\text{C}_6\text{H}_5\text{COO}^-]}{[\text{C}_6\text{H}_5\text{COOH}]} = \frac{[\text{H}^+]^2}{[0.016]}; [1 \text{ mark}]$$

- $[\text{H}^+] = \sqrt{6.3 \times 10^{-5} \times 0.016} = 1.01 \times 10^{-3}$; [1 mark]

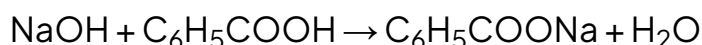
- $\text{pH} = 2.99 - 3.00$ to 2 decimal places; [1 mark]

[Total: 4 marks]

- In order to work out the pH of a weak acid, you must work out the concentration of the acid itself, $[\text{C}_6\text{H}_5\text{COOH}] / [\text{HA}]$
- Write the correct expression for $K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$
 - We can assume that the values for $[\text{H}^+]$ and $[\text{A}^-]$ are the same
 - The equation is then rearranged to give $[\text{H}^+] = \sqrt{K_a \times [\text{HA}]}$

1d

d) To calculate the pH of the solution formed:



- Moles of $\text{NaOH} = \frac{0.06}{40.0} = 0.0015 \text{ moles}$; [1 mark]

- Moles of $\text{C}_6\text{H}_5\text{COOH}$ left = $\left(\frac{0.005}{2}\right) - 0.0015 = 0.001 \text{ moles}$

AND

Moles of $\text{C}_6\text{H}_5\text{COO}^- = 0.0015$ moles; [1 mark]

Either method 1:

- $[\text{H}^+] = K_a \times \frac{[\text{acid}]}{[\text{salt}]}$; [1 mark]
- $[\text{H}^+] = 6.3 \times 10^{-5} \times \frac{0.001}{0.0015} = 4.2 \times 10^{-5}$ [1 mark]
- $\text{pH} = 4.38$ to 2 decimal places; [1 mark]

OR method 2:

- $\text{pH} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$; [1 mark]
- $\text{pH} = 4.20 + \log \frac{[0.0015]}{[0.001]}$; [1 mark]
- $\text{pH} = 4.38$ to 2 decimal places; [1 mark]

[Total: 5 marks]

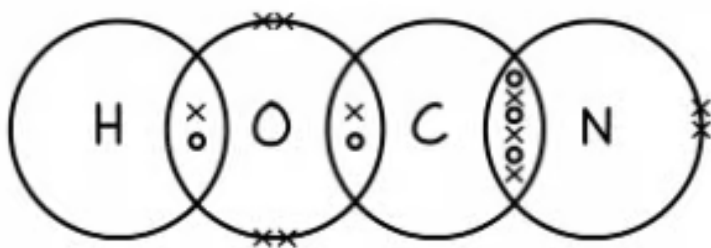
- The key with buffer calculations is to know the correct expression to use
- $K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$ where A^- represents the salt
 - This can be converted to:
 $[\text{H}^+] = K_a \times \frac{[\text{acid}]}{[\text{salt}]}$
- An alternative method is to use the Henderson-Hasselbalch equation
 - $\text{pH} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$
- To convert K_a to $\text{p}K_a$
 - $\text{p}K_a = -\log K_a$
 - So $-\log 6.3 \times 10^{-5} = 4.20$
- In this question, you will get the same answer if you use moles or concentration of the acid and salt
 - This is because to work out the concentration you can multiply the number of moles by 2 as the volume is 500.00 cm^3
 - 0.001 mols of acid becomes $0.002 \text{ mol dm}^{-3}$
 - 0.0015 mols salt becomes $0.003 \text{ mol dm}^{-3}$

2a

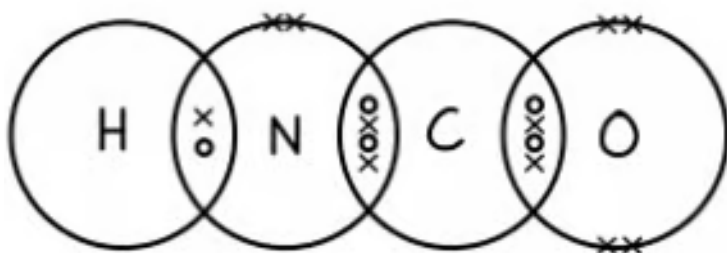
a)

i) The 'dot-and-cross' diagrams of these two acids, showing outer shell electrons

only are:



- Correct structure for HOCN; [1 mark]



- Correct structure for HNCO; [1 mark]
- 16 electrons on each diagram; [1 mark]

ii) The values of the bond angles HNC and NCO in isocyanic acid are:

- $\text{HNC} = 115 - 125^\circ$
AND
 $\text{NCO} = 180^\circ$; [1 mark]

iii) The acid with the shorter bond length is:

- Cyanic acid
AND
(Because,) it's a stronger / higher bond enthalpy / triple / $\text{C}\equiv\text{N}$ / more electrons involved bond; [1 mark]

[Total: 5 marks]

hydrogen,

- Double-check once you have drawn the diagram that there are the correct number of electrons in total
 - Carbon **4** + oxygen **6** + hydrogen **1** + nitrogen **5** = 16 electrons
- For part (ii)
 - HNC: there are three areas of electron density in the HNC bond so an acceptable answer will be in the range of 115 to 125°
 - The lone pair will affect the bond angle which is why a range is given as an answer
 - NCO: there are two areas of electron density so the bond angle will be 180 as it is linear
- For part (iii)
 - **Remember:** A triple bond is shorter than a double bond, which is shorter than a single bond

2b

b)

i) To calculate the pH

- $[H^+] = \sqrt{[HNC\bar{O}]K_a} = \sqrt{(0.1 \times 1.2 \times 10^{-4})} = 3.46 \times 10^{-3}$; [1 mark]
- $pH = -\log[H^+] = 2.5 / 2.46$; [1 mark]

ii) The equation for the formation of the sodium cyanate is:

- $Na_2CO_3 + 2(NH_2)_2CO \rightarrow 2NaNCO + CO_2 + 2NH_3 + H_2O$; [1 mark]

[Total: 2 marks]

- In order to work out the pH of a weak acid you must work out the concentration of the acid itself, $[HNC\bar{O}] / [HA]$
- Write the correct expression for $K_a = \frac{[H^+][A^-]}{[HA]}$
 - We can assume that the value for $[H^+]$ and $[A^-]$ is the same
 - $K_a = \frac{[H^+]^2}{[HA]}$
 - The equation is then rearranged
 - $[H^+] = \sqrt{K_a \times [HA]}$

2c

c)

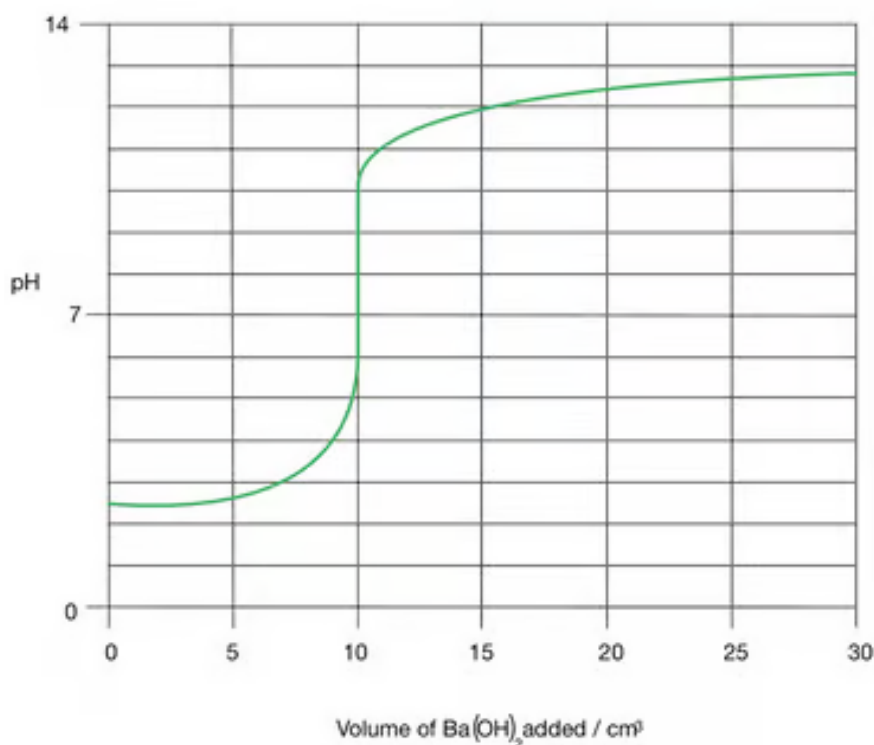
i) To calculate $[\text{OH}^-]$:

- moles of OH^- at start = $\frac{(2 \times 0.1 \times 30)}{1000} = 6 \times 10^{-3} \text{ mol}$
moles of OH^- reacted = $\frac{(0.1 \times 20)}{1000} = 2 \times 10^{-3} \text{ mol}$
moles of OH^- remaining = $(6 - 2) \times 10^{-3} = 4 \times 10^{-3} \text{ mol}$, (in 50 cm^3); [1 mark]
- so $[\text{OH}^-]_{\text{end}} = \frac{(4 \times 10^{-3} \times 1000)}{50} = 0.08 \text{ mol dm}^{-3}$; [1 mark]

ii) To calculate $[\text{H}^+]$ and the pH of the solution at the end of the addition:

- $[\text{H}^+] = \frac{K_w}{[\text{OH}^-]} = \frac{1 \times 10^{-14}}{0.08} = 1.25 \times 10^{-13} \text{ mol dm}^{-3}$; [1 mark]
- so $\text{pH} = -\log(1.25 \times 10^{-13}) = 12.9$; [1 mark]

iii) The completed graph should include:



- Curve starts at 2.46 / 2.5; [1 mark]
- Vertical portion (endpoint) at vol added = 10.0 cm^3 ; [1 mark]
- Finishes at $\text{pH} = 12.9$; [1 mark]

[Total: 7 marks]

- For part (i)
 - In order to calculate the concentration of $[\text{OH}^-]$ you must find:
 - The number of moles at the start
 - The number of moles that have reacted
 - And finally, the number of moles that remain
 - Now you have this information you can use $c = n/v$ or $c = n/v \times 1000$ to calculate the concentration
- For part (ii)
 - When you have $[\text{OH}^-]$ you can work out the pH by using
 - $[\text{H}^+] = \frac{K_w}{[\text{OH}^-]}$ and then take the $-\log$ of this answer
 - OR
 - $14 - (-\log[\text{OH}^-])$
 - Use the method that you find the most simple
- For part (iii) use vol of acid $\times$ concentration of acid = vol of base $\times$ concentration of base
 - $20 \times 0.100 = \text{vol base} \times 0.100$
 - $\frac{(20 \times 0.100)}{0.100} = 20.0 \text{ cm}^3$
 - As $\text{Ba}(\text{OH})_2$ is used, we must then divide the volume of the base by 2
 - This is because the hydroxide dissociates to release 2 moles of OH^- ions

3a

a) To calculate the pH of the buffer solution:

- Moles of acid = $\frac{0.275}{122.0} = 2.2541 \times 10^{-3}$ moles

AND

Concentration of acid = $2.2541 \times 10^{-3} \times 4 = 9.0164 \times 10^{-3} \text{ mol dm}^{-3}$; [1 mark]

- Moles of salt = $\frac{0.525}{144.0} = 3.6458 \times 10^{-3}$ moles

AND

Concentration of salt = $3.6458 \times 10^{-3} \times 4 = 0.014583 \text{ mol dm}^{-3}$; [1 mark]

- $[\text{H}^+] = K_a \frac{[\text{acid}]}{[\text{salt}]}$; [1 mark]

- $[\text{H}^+] = 6.3 \times 10^{-5} \times \frac{9.0164 \times 10^{-3}}{0.014583} = 3.90 \times 10^{-5}$; [1 mark]

- $\text{pH} = 4.41$; [1 mark]

[Total: 5 marks]

- In order to calculate the pH of the buffer solution mass in grams should be converted to concentration
- Both the acid and salt are dissolved into 250 cm^3 of water, so the concentration can be calculated by multiplying by 4 to convert to mol dm^{-3}

3b

b) To calculate the pH of the buffer solution:

- Concentration of acid = $0.2 \times \frac{130}{215} = 0.1209$

AND

$$\text{Concentration of salt} = 0.45 \times \frac{85}{215} = 0.1779; [1 \text{ mark}]$$

- $[\text{H}^+] = K_a \frac{[\text{acid}]}{[\text{salt}]}$; [1 mark]
- $[\text{H}^+] = 1.74 \times 10^{-5} \times \frac{0.1209}{0.1779} = 1.18 \times 10^{-5}$

AND

$$\text{pH} = 4.92 - 4.93; [1 \text{ mark}]$$

[Total: 3 marks]

- Careful here to use the **total** volume of both the acid and salt in order to calculate the concentrations which need to be substituted into the equation
 - The total volume of the solution is 215 cm^3

3c

c) To calculate the new pH of the buffer:

- Moles of OH^- added = $\frac{1.00}{1000} \times 1.00 = 1 \times 10^{-3}$ / 0.001 moles; [1 mark]
- Moles of acid = $0.026 - 1 \times 10^{-3} = 0.025$

$$\text{pH} = 4.92 - 4.93; [1 \text{ mark}]$$

[Total: 3 marks]

- Careful here to use the **total** volume of both the acid and salt in order to calculate the concentrations which need to be substituted into the equation
 - The total volume of the solution is 215 cm^3

3c

c) To calculate the new pH of the buffer:

- Moles of OH^- added = $\frac{1.00}{1000} \times 1.00 = 1 \times 10^{-3}$ / 0.001 moles; [1 mark]
- Moles of acid = $0.026 - 1 \times 10^{-3} = 0.025$

AND

$$\text{Moles of salt} = 0.03825 + 1 \times 10^{-3} = 0.03925; [1 \text{ mark}]$$

- Substituting into the equation:

$$[\text{H}^+] = K_a \frac{[\text{acid}]}{[\text{salt}]}$$

$$[\text{H}^+] = 1.74 \times 10^{-5} \times \frac{(0.025 / 0.0216)}{(0.03925 / 0.0216)} = 1.12 \times 10^{-5}; [1 \text{ mark}]$$

- pH = 4.95 - 4.96; [1 mark]
- The pH value of the buffer has only increased by 0.02

AND

So, the overall pH of the solution has remained constant showing the buffer has worked; [1 mark]

[Total: 5 marks]

- When a small amount of base is added to a buffer solution
 - The moles of acid decreases
 - The moles of salt increases
- Therefore, the new moles of acid and salt must be calculated and the new values can be substituted into the equation
- You can simply substitute in the new moles as you are dividing by the same volume to calculate the new concentrations of both the acid and salt
- The question has asked you to comment on the pH, which is slightly unusual, but

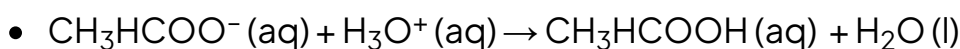
you can think about how a buffer solution is supposed to work

- If a small volume of the base is added and the buffer solution has done its job, then the pH will only change by a very small amount
- This is also why it was important to follow the instructions in the question and give your answer to two decimal places
- pH values are usually stated to two decimal places

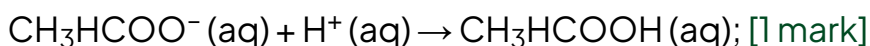
3d

d)

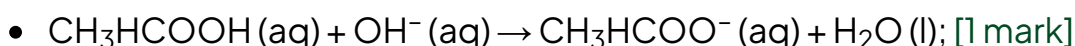
If acid is added to the buffer in part **(b)**:



OR



If base is added to a buffer in part **(b)**:



[Total: 2 marks]

- Buffers usually consist of a weak acid and its conjugate base
- When a **strong acid** (H_3O^+ or H^+) is added to a buffer solution the conjugate base present in the buffer consumes the hydronium ion converting it into water and the weak acid of the conjugate base
 - Resulting in a **decrease in the amount of conjugate base** present and an **increase in the amount of the weak acid**
- When a **strong base** (OH^-) is added to a buffer solution, the hydroxide ions are consumed by the weak acid forming water and the weaker conjugate base of the acid
- Resulting in a **decrease of the weak acid** and an **increase in the amount of the conjugate base**